



Market Data — Technical Documentation

Coresight Research Portal — Technical Documentation

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Application: Market Data Portal (MDP)

Page module: `app/pages/market_data.py` (~4,163 lines)

URL route: `/market_data`

Registered in: `app/main.py` as `st.Page("pages/market_data.py", title="Market Data", url_path="market_data")`

1. Overview

The Market Data page is the primary financial dashboard for a single selected company. It presents ten sub-tabs covering company profile, financial statements, derived metrics, analyst estimates, model forecasts, segment breakdowns, and supplemental ratings/store data. The User Interface (UI) is built to match Coresight Figma specifications: custom HyperText Markup Language (HTML) tables (not `st.dataframe`), Roboto typography, Coresight red (`#D62E2F`), sticky first columns, and client-side Excel downloads.

Data sources: Pre-ingested Alpha Vantage (AV) and Yahoo Finance (YF) tables in Azure MySQL. No live vendor Application Programming Interface (API) calls. Routing between U.S. Securities and Exchange Commission (SEC) and YFinance sources is automatic per ticker via

```
source_router.get_company_source()
```

Key design choices:

- Shared filter bar (period, dates, sort, currency, units) across most financial tabs
- Isolated filter state for **Estimates** and **Forecasting** tabs (do not overwrite shared date/period keys)
- Session-state HTML caching keyed by ticker + filter inputs (avoids re-rendering large tables on reruns)
- Cross-page ticker sync via `active_ticker` and browser `localStorage` (`sip_app_state`)
- Financial values stored in Database (DB) at **millions** scale; User Interface (UI) applies units scaling (mm / bn / k) and FX conversion

URL and deep-linking

Base URL: `/market_data`

Query parameters:

Parameter	Values	Purpose
<code>ticker</code>	Valid symbol (e.g. <code>WMT</code> , <code>ADS.DE</code>)	Selected company; validated against <code>coreiq_companies</code>
<code>tab</code>	See tab keys below	Active sub-tab; default <code>company_profile</code>
<code>period_type</code>	<code>Annual</code> <code>Quarterly</code>	Shared period type (financial tabs)
<code>start_dt</code>	ISO date <code>YYYY-MM-DD</code>	Shared range start
<code>end_dt</code>	ISO date <code>YYYY-MM-DD</code>	Shared range end
<code>sort_ord</code>	<code>Earliest</code> <code>Latest</code>	Column sort direction
<code>to_curr</code>	Currency code	Target display currency
<code>conv_mode</code>	<code>Today's Spot Rate</code> <code>Historical</code>	FX conversion mode
<code>units_sel</code>	<code>Millions (mm)</code> <code>Billions (bn)</code> <code>Thousands (k)</code>	Display units

Example deep link: `/market_data?`

`ticker=WMT&tab=income_statement&period_type=Annual&start_dt=2020-01-01&end_dt=2025-12-31`

Tab keys (`render_tabs` in `app/components/toolbar.py`):

Key	UI label	Dev badge
company_profile	Company Profile	—
key_stats	Key Stats	—
income_statement	Income Statement	—
balance_sheet	Balance Sheet	—
cash_flow	Cash Flow	—
ratios	Ratios	—
estimates	Estimates	—
forecasting	Forecasting	Testing in Progress
segment_data	Segment	Testing in Progress
ratings	Additional Data	Testing in Progress

2. Dependencies

2.1 Page entry and layout

Module	Symbols	Role
<code>components.styles</code>	<code>hide_sidebar</code> , <code>render_styles</code> , <code>set_page_layout</code> , <code>COLORS</code>	Global CSS tokens; sidebar hidden at import
<code>components.navigation</code>	<code>render_header</code> , <code>render_coresight_footer</code> , <code>render_company_header</code>	App header, company selector, footer
<code>components.toolbar</code>	<code>render_tabs</code>	10-tab navigation row
<code>components.companyProfile</code>	<code>get_company_css</code> , <code>get_profile_rows_for_excel</code> , <code>render_info_table</code> , <code>render_business_description</code> , <code>render_reference_table</code>	Company Profile tab HTML + Excel rows

2.2 Core infrastructure

Module	Symbols	Role
<code>core.auth_manager</code>	<code>require_auth</code>	Optional at page level; identity via <code>main.py</code> cookie hydrate
<code>core.database</code>	<code>db_manager</code>	Inline Structured Query Language (SQL) (period-type probe, shares/market cap for Excel)

2.3 Data layer

Module	Symbols	Role
<code>data.repository</code>	<code>CompanyOverviewRepository</code> , <code>IncomeStatementRepository</code> , <code>BalanceSheetRepository</code> , <code>CashFlowRepository</code> , <code>KeyStatsRepository</code> , <code>RatiosRepository</code> , <code>AnalystEstimatesRepository</code> , <code>ModelForecastsRepository</code> , <code>ForexRepository</code> , <code>StockQuoteRepository</code> , <code>SegmentDataRepository</code> , <code>RatingsDataRepository</code> , <code>ExecutiveCompensationRepository</code> , <code>preload_edgartools_ratings</code>	All financial data access
<code>data.source_router</code>	<code>get_company_source</code>	SEC vs YFinance routing (lazy import in page)
<code>data.models</code>	<code>IncomeStatementData</code> , <code>Company</code> , <code>BalanceSheetData</code>	Imported but unused in page logic

2.4 Utilities

Module	Symbols	Role
<code>utils.server_logger</code>	<code>log_structured_error</code> , <code>log_error</code> , <code>new_rerun_id</code> , <code>log_timing</code>	Structured logging + perf timings
<code>utils.local_storage</code>	<code>get/set_marketdata_company</code> , <code>get/set_marketdata_tab</code> , <code>get/set_marketdata_date_range</code>	Python-side persistence
<code>utils.local_storage_manager</code>	<code>sync_market_data_state</code> , <code>save_market_data_state</code> , <code>get/set_persistent_state</code>	Browser <code>localStorage</code> sync
<code>utils.ticker_utils</code>	<code>validate_and_get_ticker</code> , <code>DEFAULT_FALLBACK_TICKER</code>	Ticker validation
<code>utils.constants</code>	<code>get_currency_symbol</code> , <code>format_currency_display</code> , <code>SEGMENT_METRIC_GROUPS</code>	Currency display; segment metric order
<code>utils.excel_export</code>	<code>export_financial_excel</code> , <code>export_company_profile_simple_excel</code>	XLSX generation

2.5 Third-party

- **Streamlit** — UI, `st.cache_data`, query params, widgets
- **Plotly** (`plotly.graph_objects`, `make_subplots`) — stock price chart on Company Profile
- **concurrent.futures.ThreadPoolExecutor** — parallel DB fetches

3. UI Components

3.1 Page chrome

```
main()
  |— new_rerun_id("market_data")
  |— render_styles() + set_page_layout()
  |— render_header(current_page="market_data")
  |— render_page()
  |— render_coresight_footer()
```

`render_company_header()` (inside `render_page()`) provides:

- Page title: **CORESIGHT MARKET DATA**
- Company `selectbox` (`key="company_selector_header"`)
- **Company Documents** button → `/company_filings?ticker=...`
- Writes `active_ticker` and `selected_ticker_market_data` on change

3.2 Tab bar (`render_tabs`)

- Two-row layout: dev labels above Segment / Additional Data / Forecasting
- Active tab: red underline + bold label (HTML)
- Inactive tabs: `st.button` with `on_click` → sets `st.query_params["tab"]` + `set_marketdata_tab()`
- Horizontal rule separator below tabs

3.3 Filter row (financial tabs)

Rendered for all tabs **except** Company Profile. Hidden or simplified per tab:

Control	Widget key	Notes
Period Type	<code>period_type_select_shared</code>	Hidden for Ratings, Forecasting; disabled when only one type available
Start Date	<code>start_dt_shared</code>	Month labels from <code>available_dates</code>
End Date	<code>end_dt_shared</code>	
Sort	<code>sort_order_select_shared</code>	Earliest / Latest
Conversion	<code>conversion_mode_select</code>	Hidden for Ratios, Ratings
From Currency	<code>currency_from_unified</code>	Locked to reported currency
To Currency	<code>currency_to_unified</code>	Reported + <code>ForexRepository.get_to_currencies()</code>
Units	<code>units_select</code>	Millions / Billions / Thousands

Estimates / Forecasting: use isolated keys `_period_type_isolated_{tab}` and `_dr_isolated_{tab}_{annual|quarterly}` so filter changes do not mutate shared `date_range_market_data_shared` or URL date params.

3.4 Tables

Custom HTML `<table class="data-table">` via `st.html()` :

- Sticky first column (line item labels)
- Indent levels for subtotals (Income Statement, Balance Sheet, Cash Flow)
- Bold subtotals, grey separators, underlines per Figma (`is_bold_row` , `has_grey_separator` , `has_underline` , `get_indent_level`)
- `format_value()` applies conversion rate, units scale, comma formatting

3.5 Charts

Company Profile only: `render_stock_quote()` → Plotly line chart, ~60 months (`days=1825`) of daily close prices from `StockQuoteRepository.get_price_history()` . Left panel: quote metrics (close, open, change, 52-week H/L, market cap, shares, dividend yield).

3.6 Key Stats badges

Columns may be tagged **A** (Actual), **E** (Estimate), **F** (Forecast) with legend. Forecast columns link to `/forecasting?ticker=...`.

3.7 Excel export

`_render_excel_js_download(excel_bytes, filename, label)` — client-side Blob download via `streamlit.components.v1.html`. Bypasses Streamlit `/media/` (fails on some proxied deployments). Uses Material Symbols "table" icon and Coresight red border styling.

Per-tab builders:

- Financial tabs → `export_financial_excel()`
- Company Profile → `export_company_profile_simple_excel()` (profile + price history + shares/market cap + compensation)

3.8 Loading indicator

`_tab_loading_hint` spinner shown at start of `render_page()`; cleared after tab content renders.

4. Session State

4.1 Ticker and navigation

Key	Purpose
<code>active_ticker</code>	Cross-page company sync
<code>selected_ticker_market_data</code>	Page-local ticker
<code>selected_tab_market_data</code>	Active tab
<code>_prev_ticker_market_data</code>	Detect company change → reset filters
<code>_prev_rendered_tab</code>	Tab change logging
<code>_md_run_counter</code>	Rerun storm guard (>80 logs warning)

4.2 Shared filters

Key	Purpose
<code>date_range_market_data_shared</code>	<code>(start_iso, end_iso)</code> tuple
<code>sort_order_market_data_shared</code>	<code>Earliest</code> / <code>Latest</code>
<code>period_type_market_data_shared</code>	<code>Annual</code> / <code>Quarterly</code>
<code>target_currency</code> , <code>from_currency</code>	FX display
<code>conversion_mode</code>	Spot vs Historical
<code>units</code>	Millions / Billions / Thousands
<code>start_dt_shared</code> , <code>end_dt_shared</code>	Date widget keys
<code>sort_order_select_shared</code> , <code>period_type_select_shared</code>	Widget keys
<code>conversion_mode_select</code> , <code>currency_from_unified</code> , <code>currency_to_unified</code> , <code>units_select</code>	Widget keys

4.3 localStorage race-fix keys (`_mdlc_*`)

`_mdlc_date_range` , `_mdlc_sort` , `_mdlc_period` , `_mdlc_conv` , `_mdlc_from_curr` , `_mdlc_to_curr` , `_mdlc_units` — prevent browser hydration from overwriting in-flight widget state.

4.4 Isolated state (Estimates / Forecasting)

Pattern	Purpose
<code>_period_type_isolated_{tab}</code>	Period without affecting shared state
<code>_dr_isolated_{tab}_{annual\ quarterly}</code>	Date range per tab + period

4.5 HTML cache keys (input-hash suffixes)

Pattern	Tab
<code>_bs_html_{...}</code>	Balance Sheet
<code>_cf_html_{...}</code>	Cash Flow
<code>_seg_html_{...}</code>	Segment
<code>_ratings_html_{...}</code>	Ratings
<code>_is_html_{...}</code> (+ <code>_xl</code>)	Income Statement
<code>_ks_html_v2_{...}</code> (+ <code>_xl</code>)	Key Stats
<code>_ratios_html_{...}</code> (+ <code>_notes</code> , <code>_xl</code>)	Ratios
<code>_est_html_{...}</code>	Estimates
<code>_fcst_html_v3_{...}</code>	Forecasting

5. Data Layer

5.1 Source routing (`app/data/source_router.py`)

```
@st.cache_data(ttl=600)
def get_company_source(ticker: str) -> Optional[Literal['SEC', 'YFinance']]
```

- Reads `coreiq_companies.source`
- Composite tickers (`TSCO.L`) resolved via `exchange_acronym`
- Returns `None` if company not found

5.2 Repository classes used

Repository	Primary methods
CompanyOverviewRepository	get_company_overview(ticker)
IncomeStatementRepository	get_date_range , get_available_dates , get_income_statement_data , get_reported_currency
BalanceSheetRepository	get_date_range , get_available_dates , get_balance_sheet_data
CashFlowRepository	get_date_range , get_available_dates , get_cash_flow_data
KeyStatsRepository	get_date_range , get_available_dates , get_key_stats_data
RatiosRepository	get_date_range , get_available_dates , get_ratios_data
AnalystEstimatesRepository	get_date_range , get_available_dates , get_estimates_data
ModelForecastsRepository	get_date_range , get_available_dates , get_forecasts_data
ForexRepository	get_conversion_rate , get_to_currencies , get_conversion_rates_bulk
SegmentDataRepository	has_quarterly_segment_data , get_date_range , get_available_dates , get_segment_data
RatingsDataRepository	get_date_range , get_available_dates , get_ratings_data
StockQuoteRepository	get_latest_quote , get_price_history , get_overview_data
ExecutiveCompensationRepository	get_sec_latest_compensation , get_yf_latest_compensation

Read path: `db_manager.execute_query_readonly` (AUTOCOMMIT read engine, ~1 RTT on Azure).

SEC pattern: Wide tables with `report_type` , `fiscal_date_ending` , column-per-metric.

YFinance pattern: Long-format pivoted with `MAX(CASE WHEN line_item = ...)` grouped by `period_end` , `frequency` . Composite tickers use `yf_symbol` column.

5.3 Database tables

Table	Usage
<code>coreiq_companies</code>	Source routing; <code>name_coresight</code> for display
<code>coreiq_av_company_overview</code>	SEC company profile
<code>coreiq_yf_company_overview</code>	Yahoo Finance (YF) company profile + officers JSON
<code>coreiq_av_financials_income_statement</code>	SEC income + key stats base
<code>coreiq_yf_financials_income_statement</code>	YF income
<code>coreiq_av_financials_balance_sheet</code>	SEC balance sheet
<code>coreiq_yf_financials_balance_sheet</code>	YF balance sheet
<code>coreiq_av_financials_cash_flow</code>	SEC cash flow
<code>coreiq_yf_financials_cash_flow</code>	YF cash flow
<code>coreiq_av_financials_earnings_estimates</code>	SEC analyst estimates
<code>coreiq_yf_financials_earnings_estimates</code>	YF analyst estimates
<code>coreiq_model_forecasts</code>	Revenue model forecasts
<code>coreiq_av_forex_daily</code>	FX conversion rates
<code>coreiq_av_time_series_daily</code> / <code>coreiq_yf_time_series_daily</code>	Stock prices
<code>coreiq_av_shares_outstanding</code> / <code>coreiq_yf_shares_outstanding</code>	Shares for market cap Excel
<code>coreiq_filing_metrics_v5</code>	Segment data + ratings/store counts
<code>coreiq_executives_compensation</code>	SEC executive compensation

6. Business Logic

6.1 Ticker resolution priority

1. `active_ticker` if newer than URL (cross-page change)
2. URL `ticker` query param
3. `active_ticker` alone
4. `localStorage` stored ticker
5. `DEFAULT_FALLBACK_TICKER`

On change: reset to **5-year default** date window, Annual period, reported currency 1:1; clear legacy per-tab keys and HTML caches (new hash inputs).

6.2 Period type fallback

`get_available_period_types(ticker, tab)` probes DB for annual/quarterly availability. If user selects Quarterly but tab has no quarterly data → `st.info()` + snap to Annual.

6.3 Currency conversion

- **From:** locked to `reported_currency` (`_get_company_reported_currency` , cached 300s)
- **To:** reported + forex list
- **Spot:** single `ForexRepository.get_conversion_rate(from, to, today)`
- **Historical:** `get_conversion_rates_bulk(from, to, fiscal_dates_in_range)` applied per column

6.4 Units scaling

UI label	<code>units_scale</code>
Millions (mm)	1.0
Billions (bn)	0.001
Thousands (k)	1000.0

DB values are in millions; `format_value(value, conversion_rate, units_scale)` multiplies before display.

6.5 Tab-specific behavior

Tab	Filters	Data source	Export
Company Profile	None	Overview + quote + compensation	Profile Excel
Income Statement	Full shared	<code>IncomeStatementRepository</code>	Yes
Balance Sheet	Full shared	<code>BalanceSheetRepository</code>	Yes
Cash Flow	Full shared	<code>CashFlowRepository</code>	Yes
Key Stats	Full shared	<code>KeyStatsRepository</code> (A/E/F columns)	Yes
Ratios	Dates + sort only	<code>RatiosRepository</code> (derived)	Yes
Estimates	Isolated dates/period	<code>AnalystEstimatesRepository</code>	Yes
Forecasting	Isolated; annual only	<code>ModelForecastsRepository</code>	Yes
Segment	Full shared	<code>SegmentDataRepository</code> + <code>SEGMENT_METRIC_GROUPS</code>	Yes
Additional Data	Annual only; no FX	<code>RatingsDataRepository</code> + edgartools preload	Yes

6.6 Background preload

`preload_edgartools_ratings(ticker)` runs on every tab load (daemon thread, in-process cache). Powers ratings/store/square-footage sections without blocking UI.

7. Structured Query Language (SQL) and Query Patterns

7.1 Period type probe (in-page)

```
SELECT DISTINCT frequency -- or report_type for SEC
FROM coreiq_{yf|av}_financials_{table}
WHERE ticker = :ticker -- or yf_symbol for composite
      AND frequency IN ('annual', 'quarterly')
```

7.2 Shares + market cap (Company Profile Excel)

```
SELECT s.report_date, s.shares_outstanding_basic, s.shares_outstanding_diluted, ts.close
FROM coreiq_av_shares_outstanding s
INNER JOIN coreiq_av_time_series_daily ts
      ON ts.ticker = s.ticker AND ts.day_date = s.report_date
WHERE s.ticker = :ticker
ORDER BY s.report_date DESC
```

7.3 Repository patterns

- **Latest row per period:** `ROW_NUMBER() OVER (PARTITION BY ticker ORDER BY date DESC)`
- **Income statement optimization:** `_fetch_all_annual_rows` cached once, reused for date range + table body
- **Segment metrics:** `coreiq_filing_metrics_v5` with edgartools fallback in `SegmentDataRepository`
- **Access Control List (ACL):** Not applied in SQL on this page (no row-level user filter)

8. Caching

Layer	Mechanism	TTL
Page	<code>@st.cache_data</code> on <code>_get_company_reported_currency</code> , <code>get_available_period_types</code>	300s
Repositories	<code>@st.cache_data</code> on overview, statement fetches, key stats, ratios, estimates, ratings	300–600s
<code>source_router</code>	<code>get_company_source</code>	600s
Session state	HTML + Excel bytes keyed by inputs	Until inputs change
Module	<code>_edgar_ratings_cache</code> + preload daemon	In-process
Company map	<code>CompanyRepository.get_companies_map()</code>	Repository cache

Performance: `_timings` dict + `log_timing("PAGE_*")` summary every `render_page()` run.

9. Error Handling

Pattern	Behavior
Per-section <code>try/except</code>	<code>log_structured_error(e, page="market_data", component=..., operation=...)</code>
User messages	<code>st.error()</code> , <code>st.info("No extracted data available...")</code>
<code>main()</code> top-level	<code>log_exception</code> + re-raise
Defensive defaults	Currency → <code>"USD"</code> , conversion → <code>1.0</code> , format → <code>"-"</code>
Rerun guard	<code>_md_run_counter > 80</code> → log possible rerun storm
Best-effort	<code>preload_edgartools_ratings</code> swallows errors; exec comp section fails silently

Invalid ticker → `validate_and_get_ticker` falls back to default; `st.stop()` only if default also fails.

10. Access Control List (ACL)

Mechanism	Status
<code>require_auth(page="market_data")</code>	Optional at page level
<code>AccessControlManager.check_access("market_data", ...)</code>	Not called
<code>main.py</code> OpenID Connect (OIDC) bootstrap	Session required for background warmups; <code>pg.run()</code> unconditional

Effective access: Portal-wide OIDC session (cookie + `auth_manager.is_authenticated()`). No per-page permission levels on Market Data.

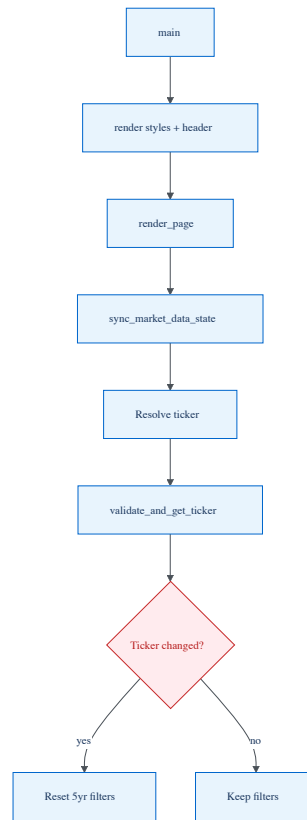
11. Flowcharts

11.1 Page load

End-to-end user journeys on this page, from load through interaction to data fetch.

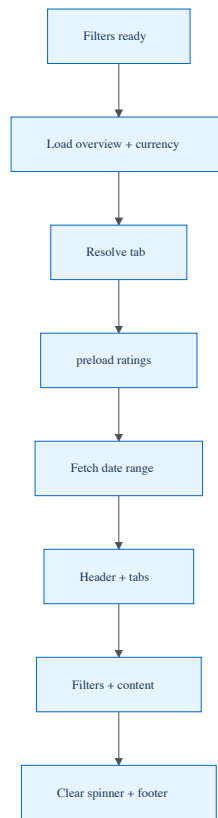
Page load — Part A: Bootstrap & ticker resolution

Page load — Part A: Bootstrap & ticker resolution



Page load — Part B: Data fetch & tab render

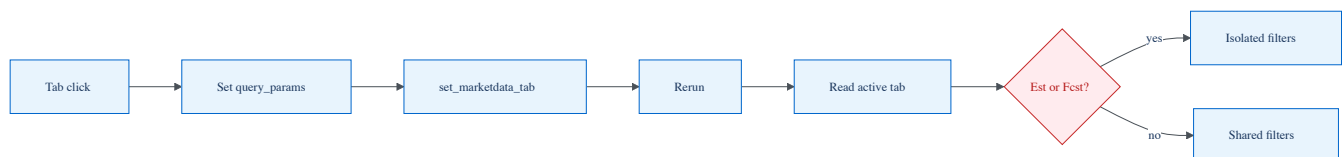
Page load — Part B: Data fetch & tab render



11.2 Tab switching

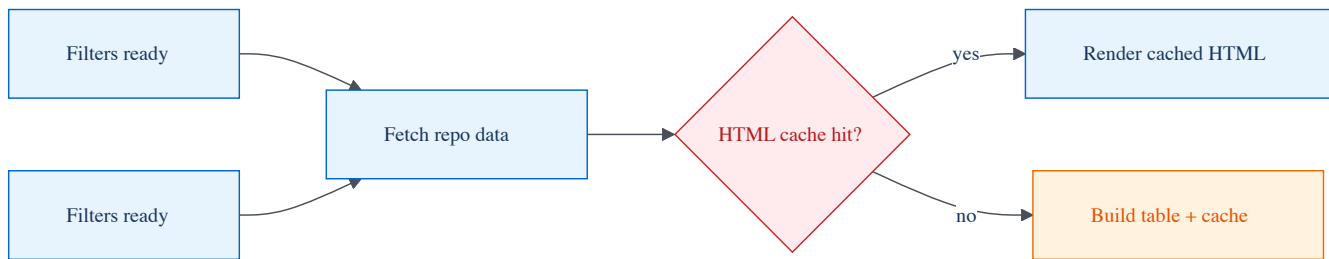
Tab switch — Part A: Navigation & filter routing

Tab switch — Part A: Navigation & filter routing



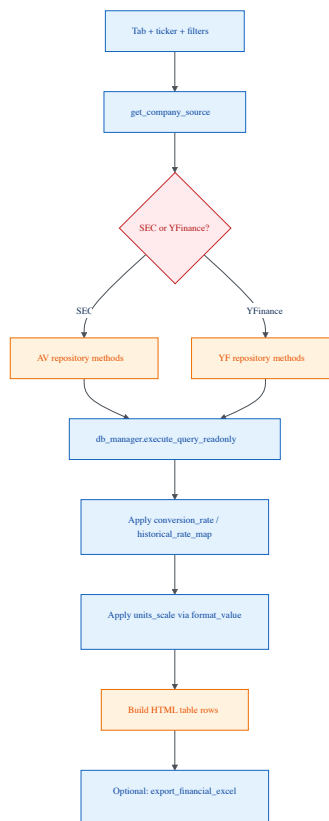
Tab switch — Part B: Data fetch & HTML cache

Tab switch — Part B: Data fetch & HTML cache



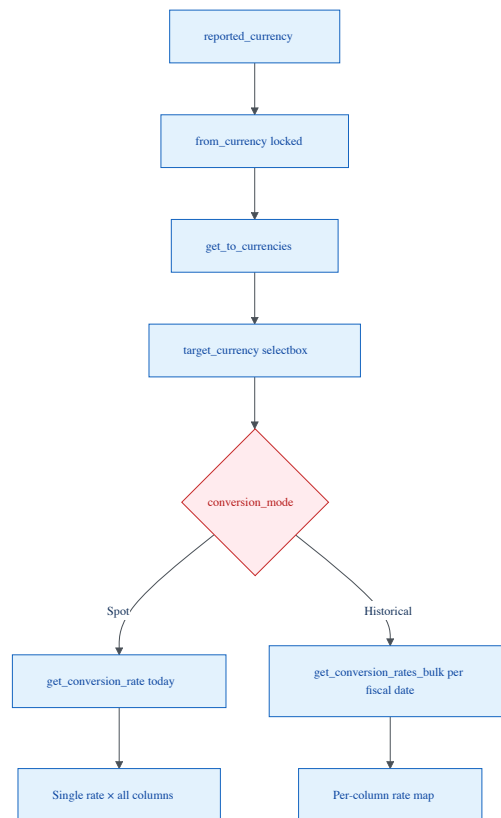
11.3 Data fetch (financial tab)

11.3 Data fetch (financial tab)



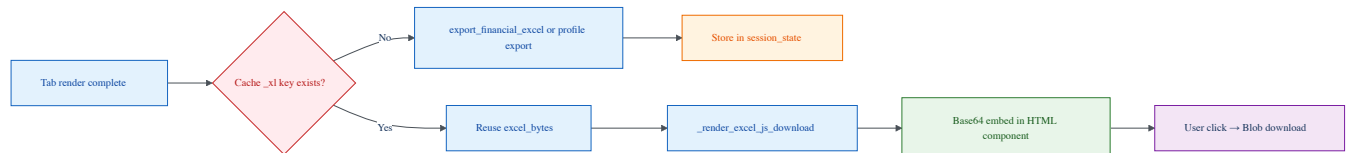
11.4 Currency conversion

11.4 Currency conversion



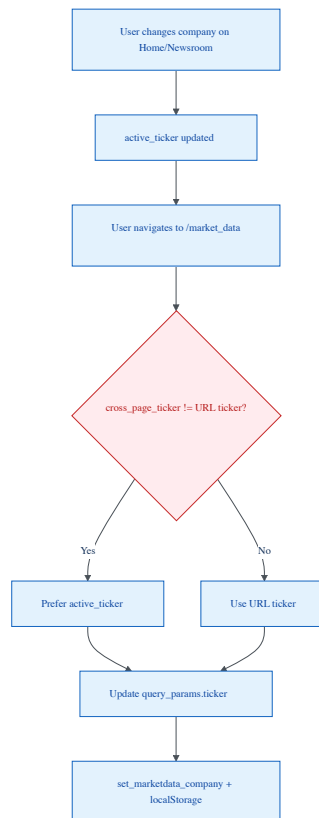
11.5 Excel export

11.5 Excel export



11.6 Cross-page ticker sync

11.6 Cross-page ticker sync



12. Feature reference by tab

12.1 Company Profile

- Info table, business description, executive compensation (SEC multi-column or YF total pay)
- Stock quote metrics + Plotly 60-month chart
- "For Reference" expander (`render_reference_table`)
- No date/currency filters

12.2 Income Statement

- Figma row styling: bold subtotals, grey separators, underlines, indent levels
- Full filter bar + Excel export

12.3 Balance Sheet / Cash Flow

- `render_balance_sheet()` / `render_cash_flow()` with indent levels 0/1/2
- Full filter bar + Excel export

12.4 Key Stats

- Combined actuals, analyst estimates, model forecast columns
- A/E/F legend; links to `/forecasting`

12.5 Ratios

- Derived from cached statements; assumption notes in `st.caption`
- No currency/units filters

12.6 Estimates

- Analyst consensus sections; EPS in reported currency; revenue converted
- Isolated filter state

12.7 Forecasting

- Ensemble + scenario bands from `coreiq_model_forecasts`
- Annual only; link to full `/forecasting` page

12.8 Segment

- Business + Geographic tables from filing metrics
- Metrics per `SEGMENT_METRIC_GROUPS` : Revenues, Operating Profit, Assets, D&A, CapEx
- Quarterly only if `has_quarterly_segment_data`

12.9 Additional Data (Ratings)

- Credit ratings (S&P, Moody's, Fitch), store counts, stores by country, square footage
- Annual only; background edgartools preload

Related pages

Page	Link
Company Filings	<code>/company_filings?ticker=...</code> (Company Documents button)
Forecasting (full)	<code>/forecasting?ticker=...</code> (from Key Stats / Forecasting tab)
Home / Newsroom	Share <code>active_ticker</code> for company sync

Generated from source analysis of `app/pages/market_data.py` and traced dependencies. Last reviewed against codebase structure as of project documentation pass.